

Git 教程

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一、Git 简介

Git 是一个使用命令提交/下载文件工具，用户可以使用命令上传本地文件到线上仓库，同时也可以使用命令将整个仓库的内容下载到本地，十分方便。

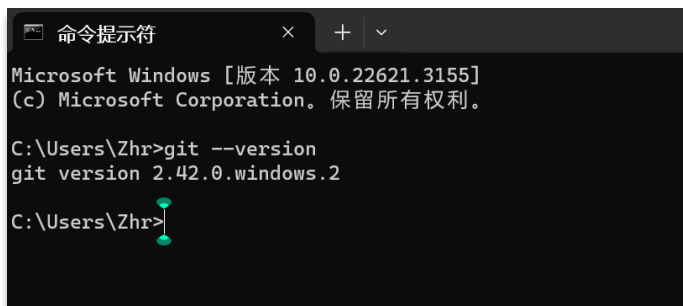
二、安装 Git

1. 安装 Git:

- (1) 官网下载安装 (<https://git-scm.com/downloads>);
- (2) 解压 git.zip 文件到 D 盘 (C 盘)。

2. 检查 Git 是否安装成功:

- (1) 同时按键 win + R, 输入 cmd 打开命令行界面, 输入 git --version :

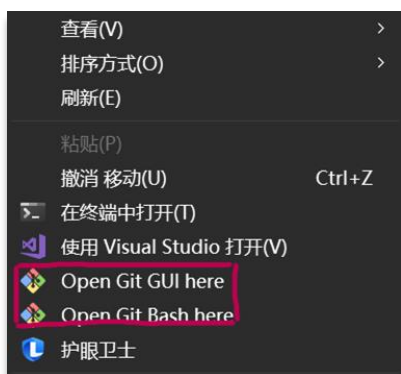


```
命令提示符
Microsoft Windows [版本 10.0.22621.3155]
(c) Microsoft Corporation。保留所有权利。

C:\Users\Zhr>git --version
git version 2.42.0.windows.2

C:\Users\Zhr>
```

- (2) 鼠标右键, 点击显示更多选项。



三、连接 Git 和 Gitee 仓库

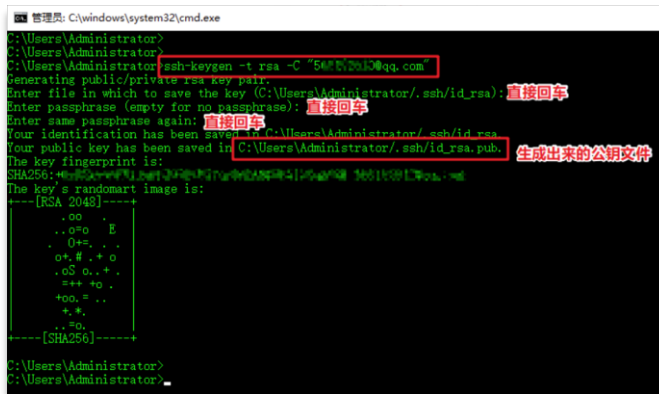
Git 和 Gitee 仓库之间传输通过 SSH 加密（SSH 是一种通信协议）需要使用 Git 生成公钥，再把公钥配置到 Gitee 上，可以理解成一把钥匙开一把锁，用 SSH 钥匙开仓库的锁。

1. 生成 SSH 公钥：

(1) 同时按键 win + R，输入 cmd 打开命令行界面，输入如下命令后回车：

ssh-keygen -t rsa -C xxxxx@xxxxx.com（下划线替换为自己的邮箱）

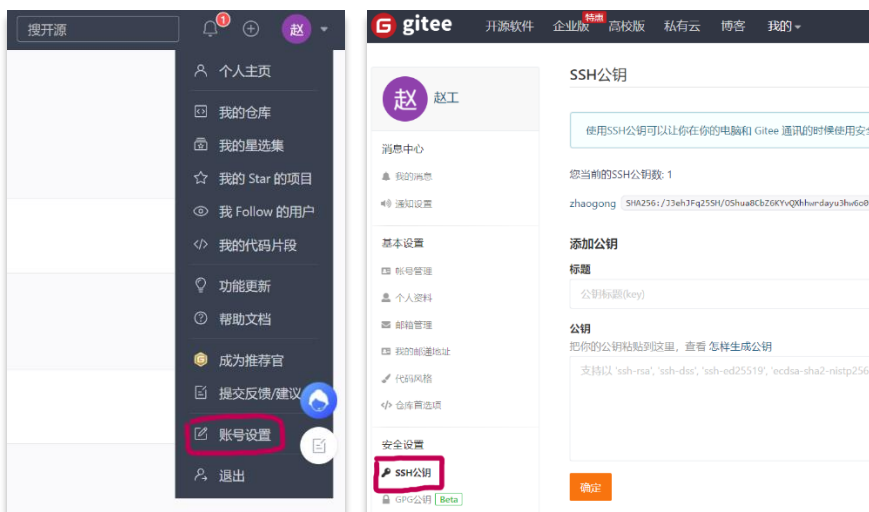
(2) 直接三次回车：



(3) 使用记事本打开文件 C:\User\Administrator\.ssh/id_rsa.pub，复制。

2. 配置 SSH 公钥到 Gitee：

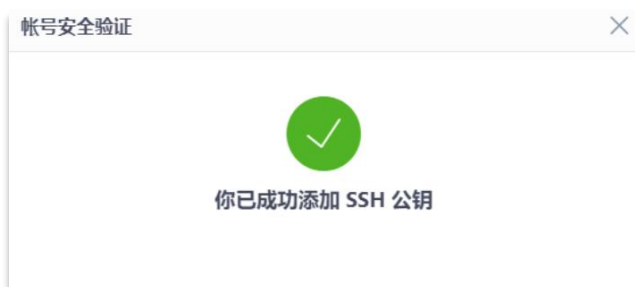
(1) 打开并登录 Gitee 一点击头像一点击账号设置—SSH 公钥



(2) 把刚刚复制的公钥粘贴过来，点击确定：



(3) 提示完成 SSH 公钥配置。



四、克隆远程仓库到本地

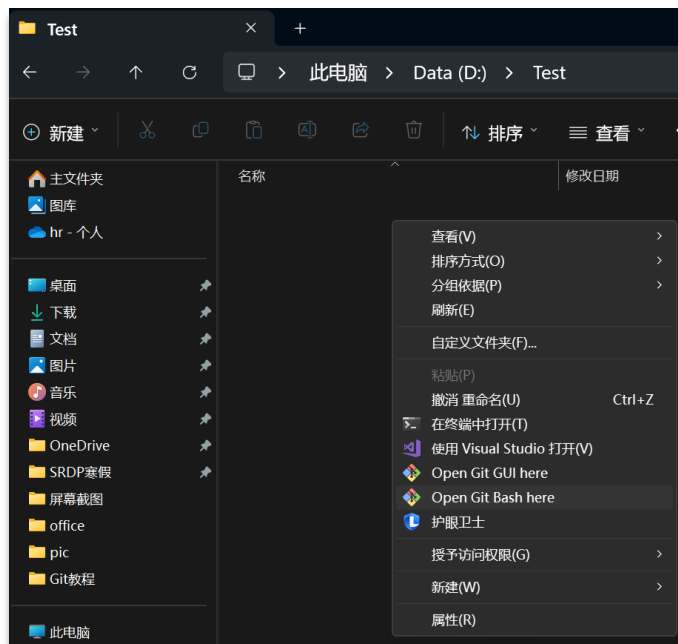
1. 打开 Test 仓库测试（两栖机器海鬣蜥项目组一仓库一Test），点击克隆：



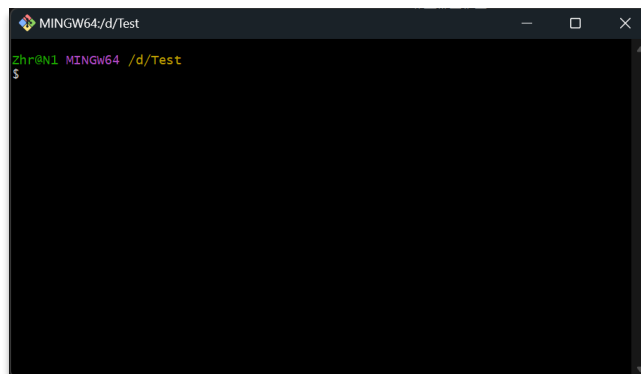
2. 点击 SSH；



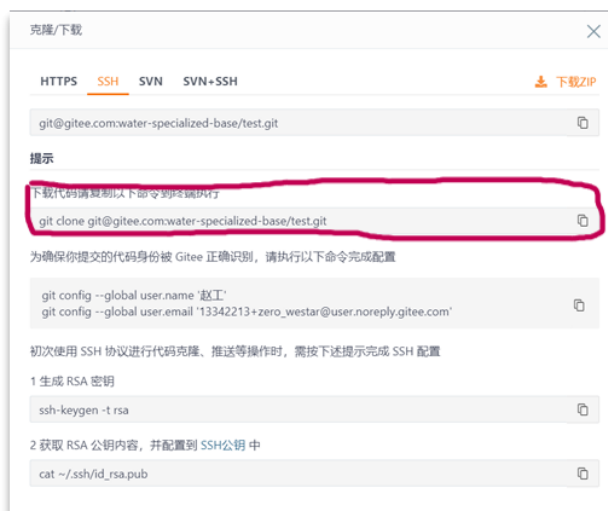
3. 在 D 盘（C 盘）创建新的文件夹“Test”——打开 Test——点击鼠标右键——点击 Open Git Bash here;



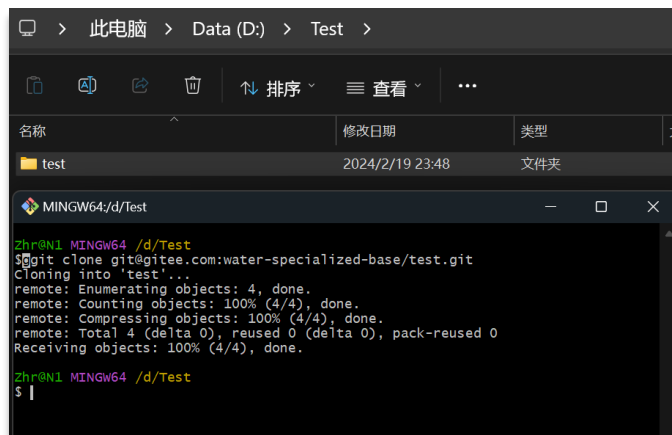
4. 弹出 Git 命令行窗口;



5. 下载仓库所有内容到本地，输入如下指令后回车;



6. 仓库成功克隆到本地文件夹。

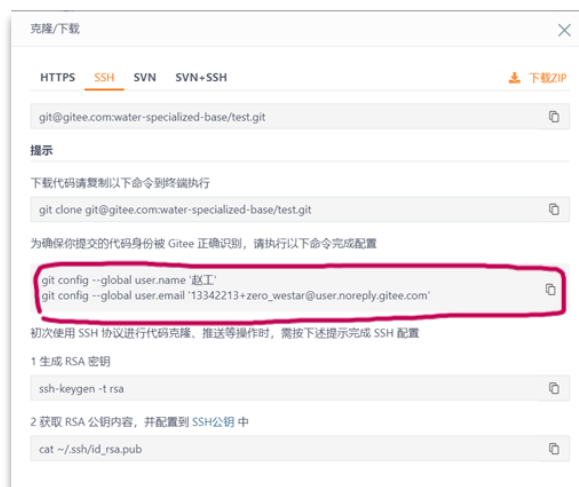


五、本地文件上传到仓库（错误示范）

1. 打开需要上传文件所在的文件夹；
2. 点击鼠标右键一点击 Open Git Bash here，弹出 Git 命令窗口；



3. Git 全局设置（设置提交者的个人信息）：输入如下命令并回车：



```
MINGW64/d/Test/上传
Zhr@N1 MINGW64 /d/Test/上传
$ git config --global user.name '赵工'
git config --global user.email '13342213+zero_westar@user.noreply.gitee.com'
Zhr@N1 MINGW64 /d/Test/上传
$
```

4. 将本地文件夹初始化为临时仓库：输入 `git init`，回车；

```
Zhr@N1 MINGW64 /d/Test
$ git init
Initialized empty Git repository in D:/Test/.git/
Zhr@N1 MINGW64 /d/Test (master)
$
```

【提示】`cd` 文件夹全称，跳转到文件夹

```
Zhr@N1 MINGW64 /d/Test (master)
$ cd 上传
Zhr@N1 MINGW64 /d/Test/上传 (master)
$
```

5. 将指定文件添加至暂存区：输入 `git add` 文件全称，回车；

```
Zhr@N1 MINGW64 /d/Test/上传 (master)
$ git add 测试文件.txt
Zhr@N1 MINGW64 /d/Test/上传 (master)
$
```

6. 将暂存区内容提交至本地仓库，输入 `git commit -m "上传测试文件"`，回车（“”中内容可改）

```
Zhr@N1 MINGW64 /d/Test/上传 (master)
$ git commit -m "上传测试文件"
[master (root-commit) 858d1f0] 上传测试文件
1 file changed, 1 insertion(+)
create mode 100644 "\344\270\212\344\274\240\346\265\213\350\257\225\346\226\207\344\273\266.txt"
Zhr@N1 MINGW64 /d/Test/上传 (master)
$
```

7. 连接远程仓库，输入：`git remote add origin` “远程仓库地址”，回车；



```
Zhr@N1 MINGW64 /d/Test/上传 (master)
$ git remote add origin git@gitee.com:water-specialized-base/test.git
Zhr@N1 MINGW64 /d/Test/上传 (master)
$
```

8. 将本地仓库 push 至远程仓库，输入：git push -u origin master ， 回车；

```
$ git push -u origin master
To gitee.com:water-specialized-base/test.git
! [rejected]        master -> master (fetch first)
error: failed to push some refs to 'gitee.com:water-specialized-base/test.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

Zhr@N1 MINGW64 /d/Test/上传 (master)
```

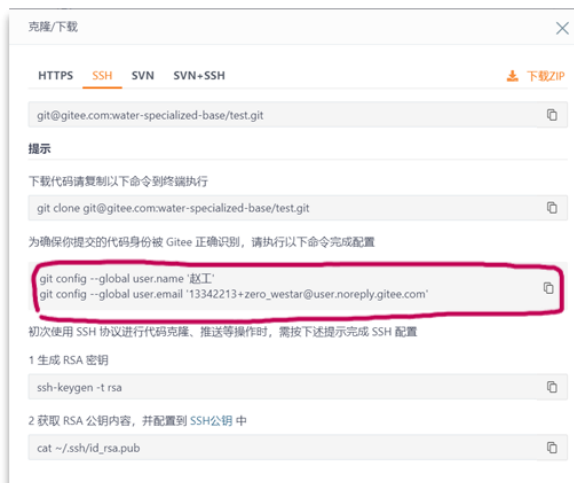
六、本地文件上传到仓库（正确示范）

1. 打开文件所在文件夹，将其初始化为临时仓库：输入 git init ， 回车；

```
Zhr@N1 MINGW64 /d/Test/上传 (master)
$ git init
Initialized empty Git repository in D:/Test/上传/.git/

Zhr@N1 MINGW64 /d/Test/上传 (master)
$
```

2. Git 全局设置（设置提交者的个人信息）：输入如下命令并回车：



```
Zhr@N1 MINGW64 /d/Test/上传 (master)
$ git config --global user.name '赵工'

Zhr@N1 MINGW64 /d/Test/上传 (master)
$ git config --global user.email '13342213+zero_westar@user.noreply.gitee.com'

Zhr@N1 MINGW64 /d/Test/上传 (master)
$
```

3. 检查当前暂存区是否为空：git status， 回车；

```
Zhr@N1 MINGW64 /d/Test/上传 (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  "\346\265\213\350\257\225\346\226\207\344\273\266.txt"

nothing added to commit but untracked files present (use "git add" to track)
```

4. 将提交文件放到暂存区: `git add` 文件全称, 回车;

```
Zhr@N1 MINGW64 /d/Test/上传 (master)
$ git add 测试文件.txt

Zhr@N1 MINGW64 /d/Test/上传 (master)
$
```

5. 再次查看暂存区发现已经添加文件: `git status`, 回车;

```
Zhr@N1 MINGW64 /d/Test/上传 (master)
$ git add 测试文件.txt

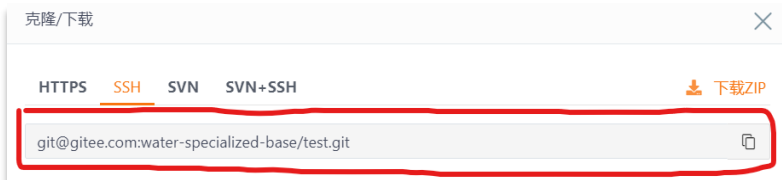
Zhr@N1 MINGW64 /d/Test/上传 (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   "\346\265\213\350\257\225\346\226\207\344\273\266.txt"
```

【提示】文件名包含中文, 6 个数字代表一个汉字

6. 远程连接到仓库: `git remote add origin` 访问路径, 回车;



```
Zhr@N1 MINGW64 /d/Test/上传 (master)
$ git remote add origin git@gitee.com:water-specialized-base/test.git

Zhr@N1 MINGW64 /d/Test/上传 (master)
$
```

7. 将远程仓库与本地仓库进行分支合并: `git pull origin master`, 回车

```
Zhr@N1 MINGW64 /d/Test/上传 (master)
$ git pull origin master
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
Unpacking objects: 100% (4/4), 2.07 KiB | 303.00 KiB/s, done.
From gitee.com:water-specialized-base/test
 * branch            master       -> FETCH_HEAD
 * [new branch]      master       -> origin/master
```

8. 把文件输出到本地仓库: `git commit -m ''`, 回车 (''中一般写更新了什么内容)

```
Zhr@N1 MINGW64 /d/Test/上传 (master)
$ git commit -m '上传测试文件'
[master f769ad4] 上传测试文件
1 file changed, 1 insertion(+)
create mode 100644 "\346\265\213\350\257\225\346\226\207\344\273\266.txt"
```


9. 把本地文件上传到仓库: git push origin master, 回车

```
zhr@N1 MINGW64 /d/Test/上传 (master)
$ git push origin master
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 20 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 362 bytes | 362.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
remote: Powered by GITEE.COM [GNK-6.4]
To gitee.com:water-specialized-base/test.git
3642db2..f769ad4 master -> master
```

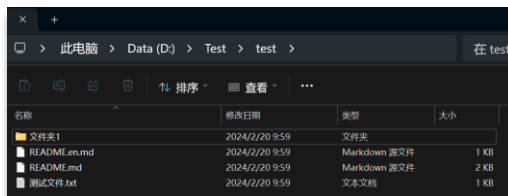
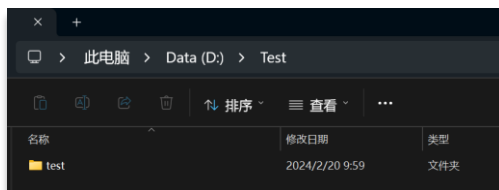
10. 打开仓库, 刷新页面, 查看上传成功。



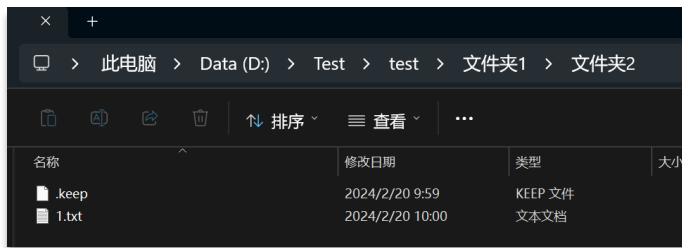
七、上传到仓库指定位置

上传文件 1.txt 到仓库位置: 文件夹 1/文件夹 2

1. 克隆远程仓库到本地, 参考 四、克隆远程仓库到本地



2. 在本地目标文件夹添加新的文件 1.txt



3. 把“文件夹1”上传到 gitee 仓库，参考“六、本地文件上传到仓库”，把文件替换成文件夹；

- (1) 六、1. 因为 test 是克隆到本地的仓库，具有本地仓库的属性，不需要 git init;
- (2) 六、2-3;

```
Zhr@N1 MINGW64 /d/Test/test (master)
$ git config --global user.name '赵工'
git config --global user.email '13342213+zero_westar@user.noreply.gitee.com'

Zhr@N1 MINGW64 /d/Test/test (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    "\346\226\207\344\273\266\345\244\2711\346\226\207\344\273\266\345\244\2712\1.txt"

nothing added to commit but untracked files present (use "git add" to track)
Zhr@N1 MINGW64 /d/Test/test (master)
$ |
```

- (3) 六、4-5;

```
Zhr@N1 MINGW64 /d/Test/test (master)
$ git add 文件夹1

Zhr@N1 MINGW64 /d/Test/test (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   "\346\226\207\344\273\266\345\244\2711\346\226\207\344\273\266\345\244\2712\1.txt"

Zhr@N1 MINGW64 /d/Test/test (master)
$
```

- (4) 六、6: 因为是从远程克隆到本地，已经连接了，可以不用执行，执行了也没事；

```
Zhr@N1 MINGW64 /d/Test/test (master)
$ git remote add origin git@gitee.com:water-specialized-base/test.git
error: remote origin already exists.
```

- (5) 六、7-8;

```
Zhr@N1 MINGW64 /d/Test/test (master)
$ git pull origin master
From gitee.com:water-specialized-base/test
 * branch      master      -> FETCH_HEAD
Already up to date.

Zhr@N1 MINGW64 /d/Test/test (master)
$ git commit -m '更新文件夹1'
[master 9fb558b] 更新文件夹1
1 file changed, 1 insertion(+)
 create mode 100644 "\346\226\207\344\273\266\345\244\2711\346\226\207\344\273\266\345\244\2712\1.txt"

Zhr@N1 MINGW64 /d/Test/test (master)
```

(6) 六、9

```
Zhr@N1 MINGW64 /d/Test/test (master)
$ git push origin master
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 20 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 457 bytes | 457.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Powered by GITEE.COM [GNK-6.4]
To gitee.com:water-specialized-base/test.git
   3736176..9fb558b  master -> master

Zhr@N1 MINGW64 /d/Test/test (master)
$
```

(7) 打开仓库查看（应该 `git commit -m '上传文件 1'` 比较合适）

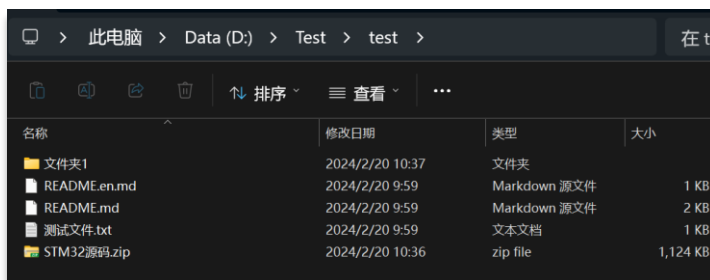


八、上传大文件

1. 下载并安装 git-lfs，官网：<https://git-lfs.github.com/>



2. 添加大文件到本地仓库



3. 开启 lfs 功能：`git lfs install`，回车；

```
Zhr@N1 MINGW64 /d/Test/test (master)
$ git lfs install
Updated Git hooks.
Git LFS initialized.

Zhr@N1 MINGW64 /d/Test/test (master)
$
```

4. 选择提交文件类型: `git lfs track "*.zip"`, 回车;

```
Zhr@N1 MINGW64 /d/Test/test (master)
$ git lfs track "*.zip"
Tracking "*.zip"

Zhr@N1 MINGW64 /d/Test/test (master)
$ |
```

5. 设置个人信息 (六、2), 连接远程仓库 (六、6) 如果执行过了可以跳过

6. 上传 `.gitattributes` 文件;

(1) 执行完上面的命令后, 会生成一个 `.gitattributes` 文件, 要将其上传到远程 gitee 仓库。
这里把 `.gitattributes` 和大文件分开上传。

(2) 分步执行

`git add .gitattributes`

`git commit -m '上传文件'`

`git push origin master` (如果提交不了, 后面可以加一个 `-f`)

```
Zhr@N1 MINGW64 /d/Test/test (master)
$ git add .gitattributes

Zhr@N1 MINGW64 /d/Test/test (master)
$ git commit -m '上传文件'
[master 343d339] 上传文件
1 file changed, 1 insertion(+)
create mode 100644 .gitattributes

Zhr@N1 MINGW64 /d/Test/test (master)
$ git push origin master
Remote "origin" does not support the Git LFS locking API. Consider disabling it
with:
  $ git config lfs.https://gitee.com/water-specialized-base/test.git/info/lfs.locksverify false
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 20 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 347 bytes | 347.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Powered by GITEE.COM [GNK-6.4]
To gitee.com:water-specialized-base/test.git
9fb558b..343d339 master -> master

Zhr@N1 MINGW64 /d/Test/test (master)
$
```

master	分支 1	标签 0	克隆/下载
赵工	上传文件 343d339	9分钟前	6次提交
文件夹1	更新文件夹1	27分钟前	
.gitattributes	上传文件	9分钟前	
README.en.md	Initial commit	11小时前	
README.md	Initial commit	11小时前	
测试文件.txt	上传测试文件	1小时前	

7. 上传大文件

(1) 分布执行:

```
git add STM32 源码.zip
git commit -m " STM32 源码文件"
git push origin master
```

(2) 出现报错

```
Zhr@N1 MINGW64 /d/Test/test (master)
$ git add STM32源码.zip

Zhr@N1 MINGW64 /d/Test/test (master)
$ git commit -m " STM32源码文件"
[master 005df97] STM32源码文件
1 file changed, 3 insertions(+)
create mode 100644 "STM32\346\272\220\347\240\201.zip"

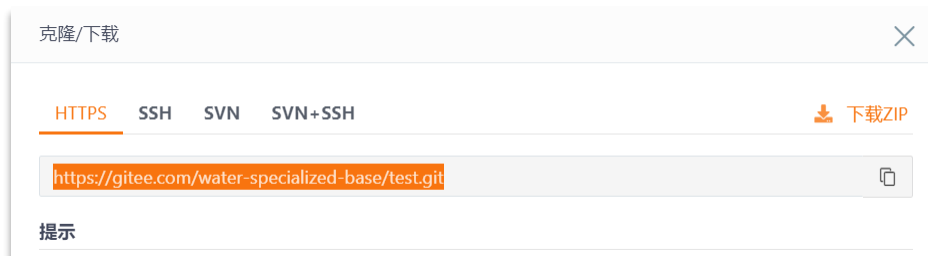
Zhr@N1 MINGW64 /d/Test/test (master)
$ git push origin master
Remote "origin" does not support the Git LFS locking API. Consider disabling it
with:
$ git config lfs.https://gitee.com/water-specialized-base/test.git/info/lfs.locksverify false
batch request: [session-7448f09d] Permission to 'water-specialized-base/test' denied
Message: LFS only supported repository in paid or trial enterprise.: exit status 1
error: failed to push some refs to 'gitee.com:water-specialized-base/test.git'
```

8. 解决报错问题

报错 1: WARNING: Authentication error: Authentication required: LFS only supported repository in paid enterprise.

解决: git config lfs.https://gitee.com/{your_gitee}/{your_repo}.git/info/lfs.locksverify false

命令中的{your_gitee}/{your_repo}是远程仓库地址（注意命令不要用错）



```
Zhr@N1 MINGW64 /d/Test/test (master)
$ git config lfs.https://gitee.com/water-specialized-base/test.git/info/lfs.locksverify false

Zhr@N1 MINGW64 /d/Test/test (master)
$ git push origin master
batch request: [session-aa43a0bd] Permission to 'water-specialized-base/test' denied
Message: LFS only supported repository in paid or trial enterprise.: exit status 1
error: failed to push some refs to 'gitee.com:water-specialized-base/test.git'
```

此时只剩一个报错

报错 2: batch response: LFS only supported repository in paid enterprise.

解决: 删除.git/hooks/pre-push 文件, 执行 rm .git/hooks/pre-push

```
Zhr@N1 MINGW64 /d/Test/test (master)
$ rm .git/hooks/pre-push

Zhr@N1 MINGW64 /d/Test/test (master)
$ git push origin master
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 20 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 437 bytes | 437.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Powered by GITEE.COM [GNK-6.4]
To gitee.com:water-specialized-base/test.git
343d339..005df97 master -> master

Zhr@N1 MINGW64 /d/Test/test (master)
$
```

9. 查看仓库, 成功上传

master	分支 1	标签 0	克隆/下载
赵工	STM32源码文件	005df97	17分钟前
文件夹1	更新文件夹1		40分钟前
.gitattributes	上传文件		22分钟前
README.en.md	Initial commit		12小时前
README.md	Initial commit		12小时前
STM32源码.zip	STM32源码文件	1.1 MB	17分钟前
测试文件.txt	上传测试文件		2小时前

九、步骤顺序错了怎么办?

1. 去 CSDN 查询撤销某一指令的命令;

- (1) git status 检查暂存区有 git add 的文件, 但是没有设置提交者信息;
- (2) 查询 CSDN, 发现 git reset HEAD 可以撤销 git add;
- (3) 执行撤销命令, 再次检查暂存区。

```
Zhr@N1 MINGW64 /d/Test/test (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   "\346\226\207\344\273\266\345\244\271\1\346\226\207\344\273\266\345\244\271\2\1.txt"

Zhr@N1 MINGW64 /d/Test/test (master)
$ git reset HEAD

Zhr@N1 MINGW64 /d/Test/test (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    "\346\226\207\344\273\266\345\244\271\1\346\226\207\344\273\266\345\244\271\2\1.txt"

nothing added to commit but untracked files present (use "git add" to track)
```

2. 删除所有文件, 重新来一遍。